

EXPERIMENT -5

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Subject Name: Database Management System

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Subject Code: 24CSH-298

Q1) A website maintains records of users who have registered on the platform and the status of confirmation requests sent to them. A confirmation request can either be successfully completed or may expire without completion.

$$\text{Confirmation Rate} = \frac{\text{Number of Successful Confirmations}}{\text{Total Confirmation Requests}}$$

For each registered user, calculate the confirmation rate, which is defined as:

If a user has never received any confirmation request, their confirmation rate should be considered 0.

Display the user ID along with the confirmation rate rounded to 2 decimal places. The result can be returned in any order.

SOFTWARE REQUIREMENTS:

Database Management System:

Oracle Database Express Edition (Oracle XE)

PostgreSQL Database

Database Administration Tool / Client Tool:

Oracle SQL Developer (for Oracle XE)

pgAdmin (for PostgreSQL)

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CODE SCREENSHOTS:

```
select s.user_id, round(Avg(
case
    when c.action= 'confirmed' then 1
    else 0
end), 2) from signups as s
left join confirmations as c
on s.user_id=c.user_id
group by s.user_id
```

OUTPUT SCREENSHOTS:

	user_id [PK] integer	round numeric
1	2	0.50
2	6	0.00
3	3	0.00
4	7	1.00

LEARNING OUTCOMES:

After completing this experiment, I was able to:

- Understand how to calculate a **confirmation rate** for each registered user.
- Calculate the **number of successful confirmations** based on confirmation status.
- Determine the **total number of confirmation requests** for each user.
- Handle users with **no confirmation requests** by assigning a confirmation rate of 0.
- Use **SQL aggregation functions** such as COUNT() and conditional counting.

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CONCLUSION:

In conclusion, calculating the confirmation rate provides a clear measure of how successfully users complete their confirmation requests. By comparing successful confirmations with the total number of requests and handling users with no requests appropriately, we can accurately determine each user's confirmation rate. Rounding the result to two decimal places ensures a clear and consistent presentation of the data.

Q2) A ride-booking company maintains information about trip requests and the users involved in those trips. Some users may be restricted from using the platform, while others are active participants. Each trip can either be completed successfully or cancelled by one of the participants.

For each day between 2013-10-01 and 2013-10-03, determine the proportion of canceled trips among all valid trips. A trip should be considered valid only when both the customer and the driver associated with that trip are active users. The final result should display the date and the calculated cancellation rate rounded to two decimal places.

We need to calculate:	But only consider trips where:	Date between
Cancellation Rate	Client is NOT banned	2013-10-01
=	AND	And
	Driver is NOT banned	2013-10-03
Cancelled Trips		

Total Valid Trips		

SOFTWARE REQUIREMENTS:

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Database Administration Tool / Client Tool:
Oracle SQL Developer (for Oracle XE)
pgAdmin (for PostgreSQL)

CODE SCREENSHOTS:

```
select t.request_at, round(Avg(case
when t.status<>'completed'
and
t.request_at>='2013-10-01'
and
t.request_at<='2013-10-03' then 1
else 0
end),2 )from Trips as t
join users as u
on t.client_id=u.users_id
join users as u2
on t.driver_id=u2.users_id
WHERE u.banned = 'No'
AND u2.banned = 'No'
group by t.request_at
```

OUTPUT SCREENSHOTS:

	request_at date	round numeric
1	2013-10-01	0.33
2	2013-10-02	0.00
3	2013-10-03	0.50

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LEARNING OUTCOMES:

After completing this experiment, I was able to:

- Understand how to calculate the **daily cancellation rate** for trips.
- Identify **valid trips** by checking whether both the client and driver are not banned.
- Apply **date-based filtering** to analyze trips within a specific time period.
- Use **conditional counting** to determine the number of cancelled and total valid trips.
- Round the calculated cancellation rate to **two decimal places** for clear presentation.

CONCLUSION:

In conclusion, this task demonstrates how to calculate the daily cancellation rate by considering only valid trips where both the client and driver are active users. By comparing cancelled trips with the total number of valid trips for each date and rounding the result to two decimal places, we can accurately measure the cancellation rate over the given period.